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Automating egg damage detection for improved quality control in the food industry using deep learning

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Abstract

The detection and classification of damage to eggs within the egg industry are of paramount importance for the production of healthy eggs. This study focuses on the automatic identification of cracks and surface damage in chicken eggs using deep learning algorithms. The goal is to enhance egg quality control in the food industry by accurately identifying eggs with physical damage, such as cracks, fractures, or other surface defects, which could compromise their quality. A total of 794 egg images were used in the study, comprising two different classes: damaged and not damaged (intact) eggs. Four different deep learning models based on convolutional neural networks were employed: GoogLeNet, Visual Geometry Group (VGG)-19, MobileNet-v2, and residual network (ResNet)-50. GoogLeNet achieved a classification accuracy of 98.73%, VGG-19 achieved 97.45%, MobileNet-v2 achieved 97.47%, and ResNet-50 achieved 96.84%. According to the results, the GoogLeNet model performed the damage detection with the highest accuracy rate (98.73%). This study encompasses artificial intelligence and deep learning techniques for the automatic detection of egg damage. The early detection of egg damage and accurate interventions highlights the significant importance of using these technologies in the food industry. This approach provides producers with the ability to detect damaged eggs more quickly and accurately, thereby minimizing product losses through timely intervention. Additionally, the use of these technologies offers a more efficient means of classifying and identifying damaged eggs compared to traditional methods.

KEYWORDS

automatic detection, deep learning, egg damage, egg quality, image classification

1 | INTRODUCTION

Eggs, known for their nutritious and satiating properties, also play a significant economic role (Waranusast et al., 2016). However, determining whether eggs are damaged poses a significant challenge for producers. The ability to identify completely broken eggshells is crucial in the food industry, as these eggs pose a higher risk of contamination, spoilage, and food safety concerns. Detecting such defects early in the production process helps reduce the likelihood of damaged eggs entering the supply chain, thereby ensuring product quality and consumer safety. Traditional methods often prove inadequate and time-consuming. Traditional methods, like manual checks and workers looking

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things over, are often slow and not very dependable. These methods depend on people's judgment, which can lead to mistakes and take time, especially in big production setups. On the other hand, deep learning algorithms can oversee the task automatically, making the process faster and more accurate. In this context, in the egg industry, it is critical to identify cracked eggs early. Swift and accurate detection of cracked eggs in egg production holds great economic and hygienic importance.

Artificial intelligence and deep learning models assist in detecting damaged eggs through image analysis (Huang et al., 2019; Thipakorn et al., 2017). On eggshells, deep learning algorithms can identify cracks and other damage, thus minimizing human errors. This technology's implementation can protect consumer health and enhance food industry quality control procedures. Artificial intelligence technologies can contribute not only to enhancing efficiency in egg production but also to more efficient resource use and reducing food waste. Deep learning models can demonstrate significant success in detecting damage in eggs by analyzing large amounts of data. Therefore, deep learning techniques being used to detect damaged eggs are important for increasing productivity and improving economic processes in agriculture.

The determination of egg damage status holds significant importance for food safety and trade. This section presents existing research on the determination of egg damage status.

In their study, Wu et al. (2018) suggested using the support vector machine (SVM) algorithm to identify eggs that have been cracked and those that are still intact. Two hundred brown chicken eggs were sampled; half of the eggs were cracked, and half were unbroken, forming two classes. Using the 10-fold cross-validation method, they were able to attain accuracy rates of 94% and 93% on the training and test sets, respectively, in this investigation (Wu et al., 2018).

According to Haoran et al. (2020), SVM algorithms are effective for detecting and analyzing egg cracks. Based on SVM, they achieved a classification accuracy of 98.75%. Approximately 200 egg images were used for the study, 100 of which were cracked eggs and 100 of which were intact eggs (Haoran et al., 2020).

Lai et al. (2021) proposed a method for detecting cracks in duck eggshells using acoustic features. In their study, they used 60 eggs and applied logistic regression to classify acoustic features at different frequency intervals, achieving a classification accuracy of 89.7% (Lai et al., 2021).

Botta et al. (2022) used convolutional neural network (CNN) for the automatic detection of cracks in eggshells. They achieved higher performance compared to SVM classifiers with the CNN model, obtaining accuracy of 98.28%, recall of 92.41%, and precision of 96.92%. The dataset used in the study comprises a total of 468 egg images, including 281 images of cracked eggs and 187 images of intact eggs (Botta et al., 2022).

For the purpose of detecting egg damage, Taspinar and Cinar (2023) examined a three-class dataset containing a total of 569 images. There are three types of data in this dataset: cracks, empty cells, and good cells. Visual Geometry Group (VGG)-16 and VGG-19 models were used to classify images using the transfer learning method using CNNs. Their classification success rate for the VGG-16 model was 93.8% and for the VGG-19 model was 95.1% (Taspinar & Cinar, 2023).

Soltani and Omid (2015) investigated the impact of physicochemical changes occurring during the storage process of poultry eggs on quality. In this investigation, a variety of machine learning techniques were used, including SVM, Naive Bayes, artificial neural networks (ANN), and decision trees. ANN achieved a success rate of 100% in predicting the freshness class for 150 intact eggs (Soltani & Omid, 2015). Vasileva et al. (2018) aimed to develop a machine vision method to analyze the external characteristics of chicken eggs. This method is grounded by calculating predefined threshold values such as geometric dimensions, shape index, and speckling degree to determine egg quality. A total of 400 egg samples were examined in the research, and the algorithm results were compared with those obtained using traditional manual methods (Vasileva et al., 2018). In the study by Junaidi et al. (2021V), the goal was to perform image classification to monitor incubator conditions using transfer learning. Among the images in the dataset are 1470 images of chicks, 1719 images of eggs, and 1715 images of incubation. They achieved a classification success rate of 90% with VGG-16 and 97% with Inception v3 (Junaidi et al., 2021V).

The purpose of this study is to improve the detection of damaged eggs throughout the egg production process using deep learning techniques. By implementing this approach, the goal is to enhance efficiency in the industry by identifying and removing damaged eggs from farms to the final consumer. Unlike previous research, this study employs a unique dataset not previously explored in literature. The dataset's diversity, encompassing eggs from various environments and types of cracks, contributes to its robustness. Four deep learning models of GoogLeNet, VGG-19, MobileNet-v2, and residual network (ResNet)-50 were applied, and their accuracy values and training times were compared.

There are two sections in the study, the second covers relevant literature, and the third covers methods, metrics, and datasets. An analysis of the experimental results is presented in the fourth section. Finally, the study's conclusions and recommendations are discussed in the fifth section.

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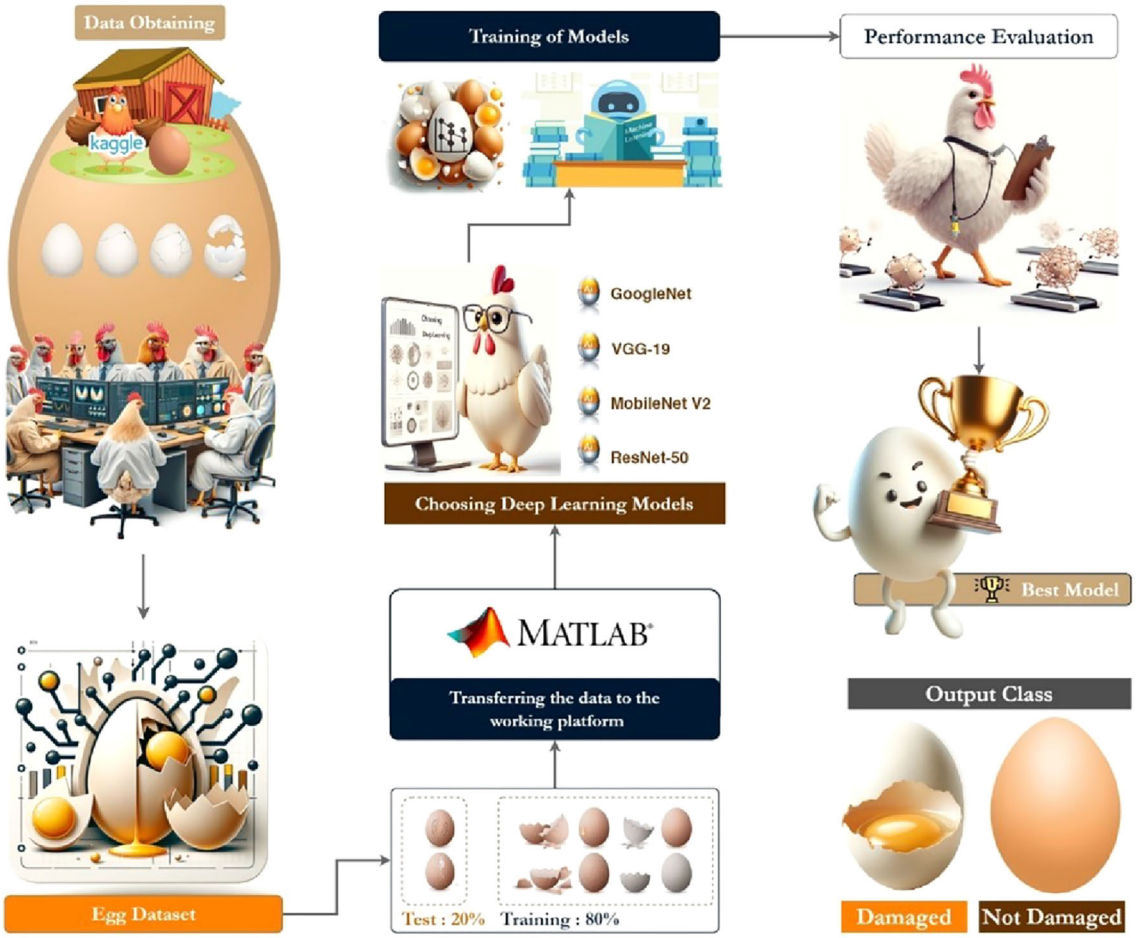


FIGURE 1 Flow diagram for the egg quality classification. *Source*: (draw.io, 2021; Payne, 2021).

2 | MATERIALS AND METHODS

In this study, a total of 794 egg images were processed, and deep learning algorithms were used for the automatic detection of any damage to the eggs. The deep learning methods used were GoogLeNet, VGG-19, MobileNet-v2, and ResNet-50, respectively. The approach followed in the study is illustrated in Figure 1. The dataset was obtained from the public repository of Kaggle, after the dataset split into train and test data. The deep neural networks were applied for this dataset, four various models applied then the outcomes were evaluated.

2.1 | Dataset and data splitting

The dataset used in this study was obtained from a public repository that contains 794 images of eggs, divided into two categories: damaged and not damaged. Each image has dimensions of 960 × 1280 pixels, a horizontal and vertical resolution of 96 dpi, and a color depth of 24 bits. The images were taken in diverse real-world environments such as kitchens, farms, markets, streets, and gardens, with varying backgrounds, including sand, grass, carpets, metal surfaces, and plastic bags. This diversity in locations and conditions ensures that the dataset captures a wide range of real-life scenarios where eggs are handled and inspected, enhancing the robustness and applicability of the model.

The "damaged" category consists of 632 images of eggs showing various forms of damage. Some eggs are cracked from one side, others are severely broken into multiple small pieces, whereas others show cracks caused by human handling during food preparation or accidental damage in different environments. Additionally, some damaged eggs are boiled, where the cracking occurs after the eggs are cooked. These variations in damage types help the model detect different severities and forms of egg damage.

The "not damaged" category includes 162 images of intact eggs, taken in different settings and under various conditions similar to those of the damaged eggs. These images depict eggs placed in refrigerators, on sand, grass, or metal, in pots, all from different angles. The diversity of the intact egg images ensures that the model is not biased

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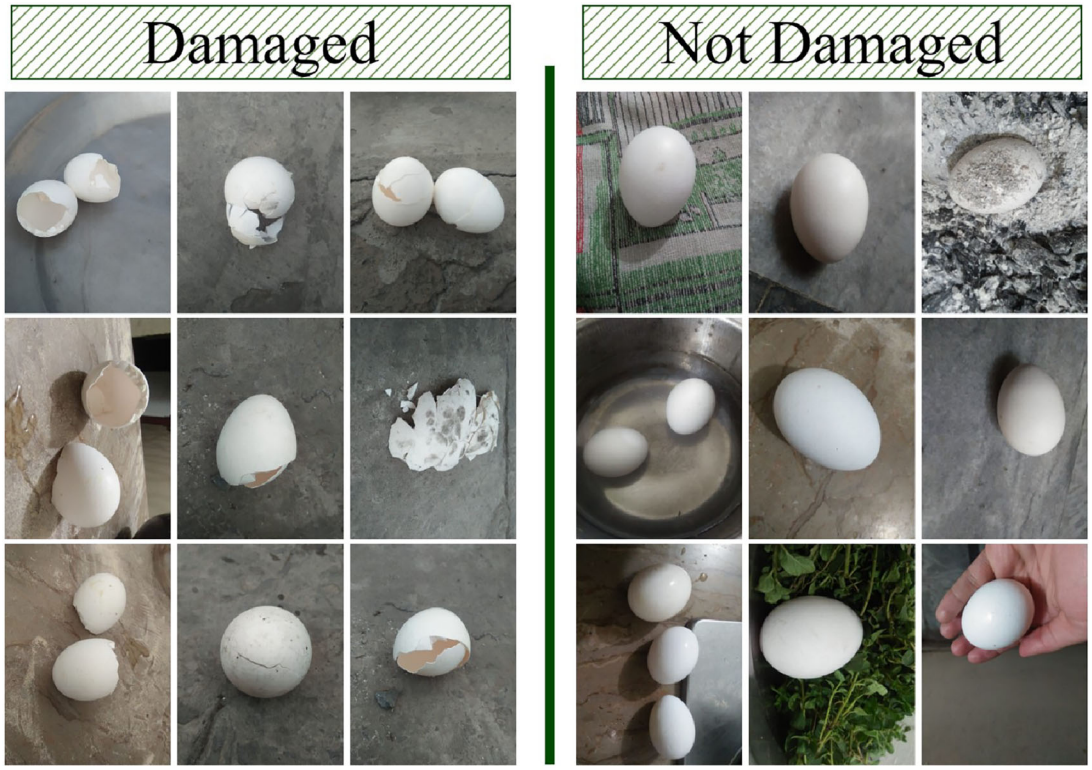


FIGURE 2 Sample images for egg dataset.

TABLE 1 Distribution of images in the egg dataset.

Feature	Number of data
Damaged	632
Not damaged	162

by specific background conditions or perspectives (Kakar, 2024). Table 1 presents the counts of damaged and not damaged images, and Figure 2 provides representative sample images from both categories.

2.2 | Convolutional neural network (CNN)

CNN, an important ANN architecture in the field of deep learning, is particularly effective in tasks involving visual data processing (Ayadi et al., 2021). It automatically classifies and analyzes input images in applications such as image recognition, object detection, and facial recognition. Unlike traditional ANNs, CNN enables the automatic learning of features in a hierarchical manner. It consists of convolutional and pooling layers that emphasize these features, allowing for the detection of complex and abstract features. CNN achieves high success rates in visual data analysis (Hussain et al., 2019). The CNN architecture is illustrated in Figure 3.

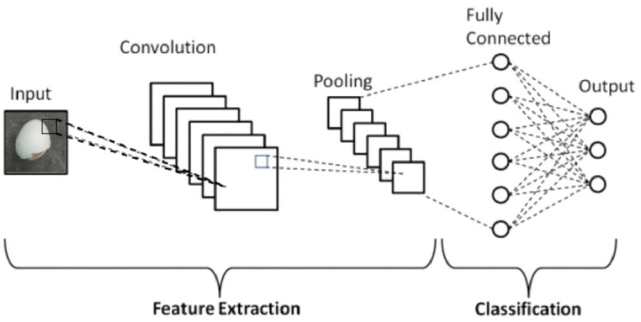


FIGURE 3 Convolutional neural networks (CNN) architecture. *Source*: Gurucharan (2024).

The convolutional layer progresses over the image with its included filters to create feature maps. The generated feature maps are matrices containing positive and negative pixel values. The subsequent activation layer, known as rectified linear unit (ReLU), utilizes an activation function. ReLU forwards positive values as output to the next layer, whereas it outputs zero for negative input values. Through the activation function, features are simplified, and the computational time of the model is reduced. The data from the ReLU layer is taken as input to the pooling layer, which reduces the input size through this layer. The fully connected layer, the last layer of the CNN architecture, is a type of neural network where classification processes take place (Al-doorı et al., 2021).

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2.2.1 | GoogLeNet

GoogLeNet is a deep learning model developed by Google. A notable feature of GoogLeNet is its use of a specific structure called the "Inception" architecture (Cinar, 2023; Yasin, Kursun et al., 2023). The inception architecture increases both the depth and width of the network by using convolutional filters and pooling operations of different sizes in parallel. This makes the network more efficient while reducing computational costs. GoogLeNet has the capacity to classify approximately 1000 image patterns. Moreover, it uses approximately 12 times fewer parameters compared to AlexNet (Al-Huseiny & Sajit, 2021). As GoogLeNet uses 1×1 convolutions, there are fewer parameters, thereby requiring less memory and computing power. The pretrained GoogLeNet model is a DAGNetwork with 144 layers and 170 connections. It is a 22-layer deep CNN with approximately 7 million parameters. The network accepts input images of $224 \times 224 \times 3$ (RGB).

2.2.2 | VGG-19

At the University of Oxford, the VGG has developed a deep learning model called VGG-19. This model was developed specifically for the recognition and classification of images. There are 19 layers in the VGG-19 deep neural network. In addition to convolutional layers, pooling layers, and fully connected layers are also included in these layers. Due to being trained on large datasets such as ImageNet, the VGG-19 model has gained the ability to recognize many different objects and scenes (Wen et al., 2019). The pretrained VGG-19 model is a SeriesNetwork with 47 layers and 46 connections. It has a depth of 19 layers and consists of approximately 144 million parameters.

The design consists of two convolutional layers followed by a max-pooling operation, and the input picture size is $224 \times 224 \times 3$. One max-pooling layer and two further convolutional layers come after this. Three further convolutional layers, one max-pooling layer, three convolutional layers, and one max-pooling layer follow these layers. The fully connected layer receives the data after these layers' feature extraction and reduction processes. Then, ReLU is employed in VGG-19 as the activation function, same like in VGG-16 (Theckedath & Sedamkar, 2020).

2.2.3 | MobileNet-v2

A lightweight and quick CNN architecture is called MobileNet-v2. MobileNet-v2's primary objective is to deliver excellent accuracy with minimal memory, processing, and computational resource requirements (Zaki et al., 2020). MobileNet-v2 is typically constructed using convolutional and depth wise separable convolution layers. These layers reduce the weights and computational costs, resulting in a lighter model (Kaya & Gürsoy, 2023). The pretrained MobileNetV2 model is a DAGNetwork with 154 layers and 163 connections. It has a depth of 53 layers and approximately 3.5 million parameters. Object identification, facial recognition, and other visual recognition tasks are frequently performed with MobileNet-v2.

2.2.4 | ResNet-50

ResNet-50 is another CNN-based model consisting of 48 convolutional layers (Mascarenhas & Agarwal, 2021). ResNet-50 is a ResNet model comprising 50 layers. This model is a variation of ResNet architecture and has gained significant traction in the field of deep learning. The key feature of ResNet-50 is its inclusion of special connections called "skip connections" or "shortcut connections" along with three 3×3 convolutional layers in each block. These connections are used to prevent the degradation of performance as the network deepens (Gencturk et al., 2024; Isik et al., 2024; Sarwinda et al., 2021). ResNet-50 is commonly used in visual recognition tasks such as image classification and object recognition. Additionally, it is often used as pretrained models with transfer learning methods. The pretrained ResNet-50 model is a DAGNetwork with 177 layers and 182 connections. It is a 50-layer deep CNN with approximately 25.6 million parameters.

2.3 | Confusion matrix and performance metrics

The confusion matrix is a tool used to evaluate the performance of classification models, helping to identify errors and understand how well the model is predicting across different classes (Al-doorı et al., 2021; Yasin, Ozkan et al., 2023). In this study, the confusion matrix is important for checking how well our deep learning models can tell if eggs are damaged or intact. It shows us where the models are doing well and where they might be making mistakes, especially as the dataset has more damaged eggs than intact ones (Yasar et al., 2019). The representation of the confusion matrix is shown in Figure 4.

There are four primary terms in the confusion matrix:

The number of instances in which the model accurately predicts the "damaged" class is known as true positive (TP).

True negative (TN): The quantity of instances in which the model accurately forecasts the "not damaged" category.

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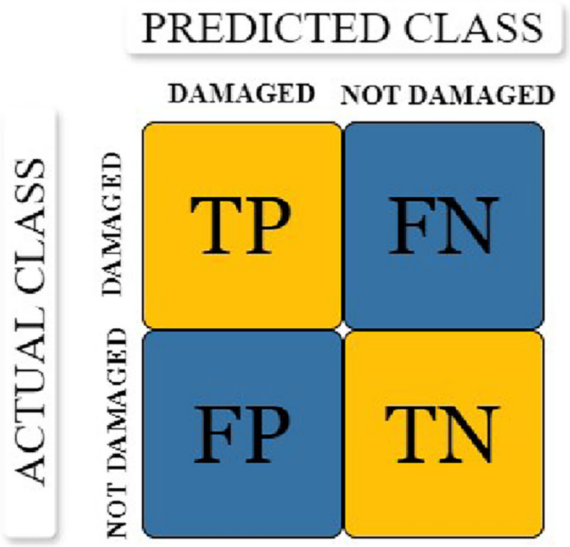


FIGURE 4 Confusion matrix for binary class.

False positive (FP): The quantity of instances in which the model predicts "damaged" (but in reality, "not damaged") wrongly.

False negative (FN): The quantity of instances in which the model predicts "not damaged" (while in fact it says "damaged").

The confusion matrix is especially useful for analyzing class imbalance impact. By examining the numbers of TP, TN, FP, and FN, we can calculate additional metrics such as precision, recall, and F1-score, which provide deeper insights into the model's ability to detect damaged eggs (the minority class). These metrics are essential, as high accuracy alone may not capture the model's performance effectively, especially when one class (damaged eggs) is underrepresented. Therefore, the confusion matrix not only allows us to measure how well the models predict but also guides further improvements by highlighting areas where errors, such as FPs or false negatives, are most common.

To provide a more comprehensive evaluation, additional metrics have been taken into consideration with accuracy:

Precision: Calculates the percentage of actual positive results among all projected positive results.

Recall: Calculates the percentage of real positives that are true positives.

F1-score: A harmonic means of precision and recall, which balances both metrics and gives a more holistic view of model performance, especially for imbalanced datasets.

AUC: Used to evaluate the performance of a classification model, representing the model's ability to distinguish between classes, with a higher AUC indicating better performance.

The performance metrics formula is presented in Table 2 (Tutuncu et al., 2022).

TABLE 2 Performance metrics formulas.

Performance metrics	Formulas
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	$\frac{TP}{TP+FP}$
Recall	$\frac{TP}{TP+FN}$
F1-score	$2\times\frac{Precision\times Recall}{Precision+Recall}$

2.4 | Experimental setup

In the process of model creation and calculations, MATLAB (MATLAB R2021b, MathWorks Inc., Deep Learning Toolbox) application was applied. For all models used in the study, the dataset was split into 80% training and 20% testing sets. Classification was performed using GoogLeNet, VGG-19, MobileNet-v2, and ResNet-50 deep learning models, respectively. The GoogLeNet model used in the study consists of 144 layers and 170 connections, the VGG-19 model made of 47 layers and 46 connection layers, the MobileNet-v2 architecture consists of 154 layers and 162 connections, and the ResNet-50 model consists of 177 layers and 192 connections.

The hardware configuration for model training included high-performance components. 64 GB of RAM, an Intel I7-12700K processor running at 3.61 GHz, and an NVIDIA RTX 3080 Ti graphics processing unit (GPU) were used to train all the models employed in the study.

Table 3 presents the common training parameters for all models.

The initial learning rate of 0.001 was chosen to strike a balance between convergence speed and stability. A lower learning rate could slow down the learning process, whereas a higher learning rate could cause the model to overshoot the optimal weights. This value is a commonly used starting point that ensures the model converges smoothly. The SGDM (stochastic gradient descent with momentum) optimizer selected because it helps accelerate convergence by adding momentum to the gradient updates, which is particularly useful for deep architectures like VGG-19 and ResNet-50. This momentum prevents the optimizer from getting stuck in local minimum and accelerates the learning process. The validation frequency of five was chosen to strike a balance between checking model performance and training time. By validating the model every five iterations, we can monitor its progress without significantly interrupting the training flow. The mini-batch size of 11 was selected to optimize GPU memory usage while still allowing for frequent updates to the model's weights. A smaller mini-batch size allows the model to update weights more often, which can lead to faster convergence but may introduce some variability in the training process. Lastly, eight epochs were used to ensure the models had enough iterations to learn

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TABLE 3 Training parameters for all models.

Initial learn rate	Solver	Validation frequency	Mini batch size	Max epochs
0.001	SGDM	5	11	8

Abbreviations: SGDM, stochastic gradient descent with momentum.

TABLE 4 Accuracy percentage and training time values of the models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	Training time (s)
GoogLeNet	98.73	98.41	100	99.2	0.9412	556
VGG-19	97.45	100	96.92	98.44	0.9846	517
MobileNet-v2	97.47	96.83	100	98.39	0.8889	443
ResNet-50	96.84	96.03	100	97.98	0.8649	456

effectively without risking overfitting. This moderate number of epochs allowed for good performance without excessively long training times.

3 | EXPERIMENTAL RESULTS

The pretrained deep learning models of GoogLeNet, VGG-19, MobileNet-v2, and ResNet-50 were trained to classify eggs as either damaged or not damaged. The models were trained, and their respective accuracy levels were recorded. Among them, GoogLeNet achieved the highest accuracy at 98.73%, followed by MobileNet-v2 and VGG-19 with 97.47% and 97.45%, respectively. ResNet-50 had the lowest accuracy, at 96.84%. These results, along with the training accuracy graph, illustrate the effectiveness of each model in identifying damaged eggs. Figure 5 provides the training accuracy graph.

The classification accuracy and training times of the deep learning models are shown in Table 4.

GoogLeNet showed a precision of 98.41%, a recall of 100%, an F1-score of 99.2%, and AUC of 0.9412 making it the best-performing model across these metrics with the longest training time. This suggests that GoogLeNet not only had high accuracy but also demonstrated excellent precision and recall in detecting damaged eggs. VGG-19 achieved a precision of 100% and a recall of 96.92%, an AUC 0.9846, with an F1-score of 98.44%. Although it had perfect precision, its recall was slightly lower than that of GoogLeNet, indicating that it may have missed some damaged eggs, though overall performance remains strong. MobileNet-v2 had a precision of 96.83%, with a recall of 100%, an AUC 0.8889, and an F1-score of 98.39%. Despite having slightly lower precision, it maintained perfect recall, suggesting it identified all damaged eggs, but at the cost of misclassifying some intact eggs. ResNet-50 had a precision of 96.03%, a recall of 100%, an AUC 0.8649, and an F1-score of 97.98%. Although it performed slightly lower than MobileNet-v2 in terms of precision and F1-score, it still achieved perfect recall, making it reliable for identifying all damaged eggs.

Upon examining Table 4, it is observed that the GoogLeNet model, which has the highest classification accuracy, also has the longest training time (556 s) compared to other models. The fastest model is observed to be MobileNet-v2 with a training time of 443 s. Training times may vary depending on the complexity and depth of the model; more complex models may require longer training times, whereas lighter models can be trained faster.

Following the training of the dataset, the confusion matrix results generated based on the learning performances of four different deep learning models (GoogLeNet, VGG-19, MobileNet-v2, ResNet-50) are presented in Figure 6.

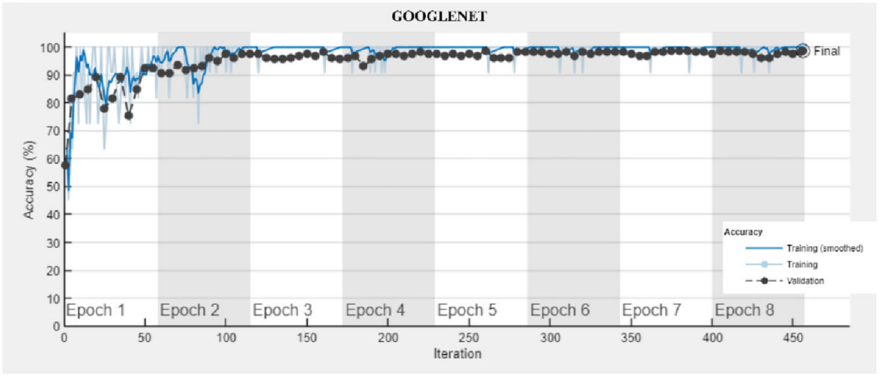
Upon analyzing Figure 6, it is observed that the most successful model, GoogLeNet, correctly classified a total of 165 instances on the test dataset, consisting of 124 damaged eggs and 32 intact eggs. The VGG-19 model, with the same accuracy value, correctly classified a total of 154 instances, including 126 damaged eggs and 28 intact eggs. Similarly, the MobileNet-v2 model, also with the same accuracy value, correctly classified 154 instances, including 122 damaged eggs and 32 intact eggs. The ResNet-50 model, with the lowest accuracy value, correctly classified a total of 151 instances, consisting of 121 damaged eggs and 30 intact eggs.

When examining the number of instances falsely labeled not damaged (FN), it is observed that only the VGG-19 model made 4 incorrect predictions. Regarding the number of instances falsely labeled damaged (FP), it is observed that GoogLeNet, MobileNet-v2, and ResNet-50 made 2, 4, and 5 incorrect predictions, respectively.

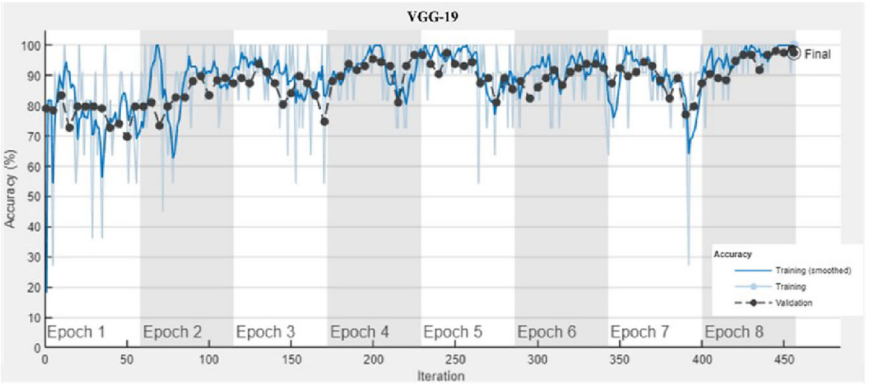
4 | DISCUSSION AND CONCLUSION

Eggs are consumed as a staple food worldwide. Throughout the production-to-consumption process, eggs must be

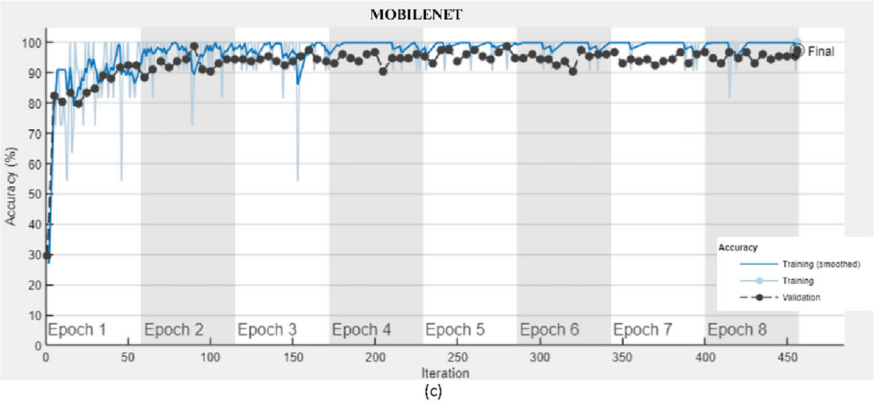
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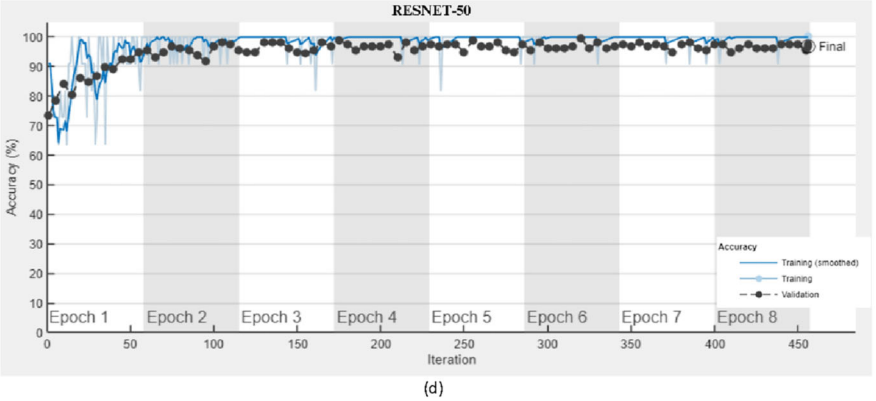
(a)



(b)



(c)



(d)

FIGURE 5 Training and verification graphs for all models: (a) GoogLeNet, (b) VGG-19, (c) MobileNet-v2, and (d) ResNet-50.

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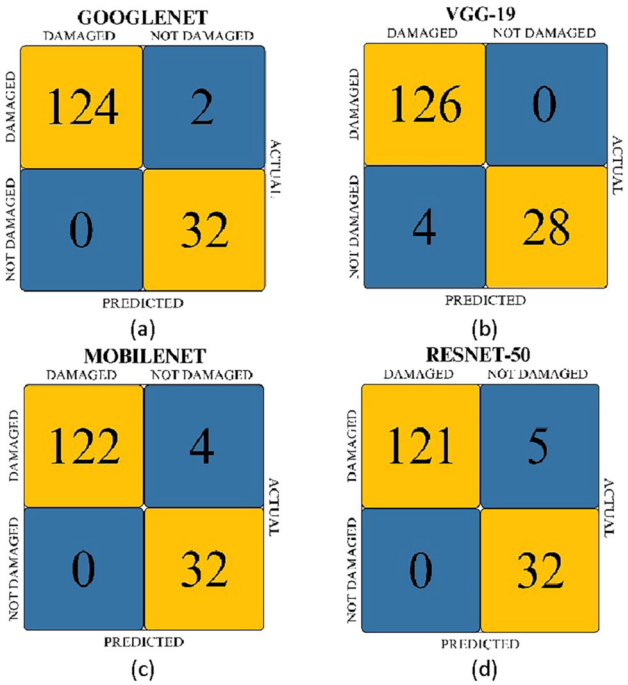


FIGURE 6 Confusion matrix results for all models (a) GoogLeNet, (b) VGG-19, (c) MobileNet-v2, and (d) ResNet-50.

protected against problems such as breakage and cracking. This situation leads to both food waste and consumer dissatisfaction. In this study, damage detection in egg images is performed using deep learning algorithms. For this purpose, the performances of GoogLeNet, VGG-19, MobileNet-v2, and ResNet-50 deep learning models are compared. The dataset used in the study consists of two classes: intact and damaged eggs. The dataset is compiled from real egg images and comprises a total of 794 egg images, including 632 damaged and 162 intact ones. According to the results obtained in the study, it is observed that the GoogLeNet model exhibits the best performance in detecting damaged eggs. Although the accuracy rate of the GoogLeNet model is 98.73%, the accuracy rates of the VGG-19, MobileNet-v2, and ResNet-50 models are 97.45%, 97.47%, and 96.84%, respectively.

Considering the mentioned metrics in the results section, GoogLeNet stands out as the best model overall due to its high accuracy, precision, recall, and F1-score. However, depending on the specific application, other models like MobileNet-v2 and ResNet-50 may be preferred for their faster training times (443 s for MobileNet-v2 and 456 s for ResNet-50), despite slightly lower precision or F1-scores.

The findings of this study indicate that deep learning algorithms can be an effective tool for detecting damage in egg images. Models with high accuracy rates, such as GoogLeNet, can be used for automatic detection of cracked or damaged eggs in egg transportation and the food industry. This can contribute to reducing food waste and increasing consumer satisfaction. In future research, the performance of deep learning models can be further enhanced by using larger and more diverse datasets. Additionally, different deep learning architectures and algorithms can also be explored.

In the future, advancements in image recognition algorithms, coupled with the integration of thermal imaging technology, hold great promise for improving the detection of smaller damage, including micro-cracks, on the surface of eggs. These innovations will enhance the ability to identify subtle defects that may not be visible with conventional imaging techniques. As a result, producers will be able to more effectively separate less damaged eggs from those with more severe cracks or defects, thereby optimizing the sorting and quality control processes. Furthermore, the use of thermal imaging will enable producers to detect damage that is not immediately visible on the surface, further enhancing the accuracy and reliability of the detection system. As deep learning models continue to be refined and trained with increasingly larger and more diverse datasets, we can expect significant improvements in the accuracy rates of systems used for detecting damaged eggs. Larger datasets will help the models learn more robust features and handle variations in egg appearances more effectively. Additionally, the application of data augmentation techniques could be used to balance the underrepresented class of intact eggs. This will help address the issue of class imbalance in the dataset, ensuring that the model has equal exposure to both classes during training. Data augmentation will create synthetic samples of the minority class (intact eggs), thus improving model generalization and performance.

Moreover, exploring other image preprocessing and processing techniques—such as advanced image filtering, edge detection, and noise reduction—could further optimize the input data for deep learning models. Techniques like contrast enhancement or histogram equalization could improve image quality, making the critical features of both intact and damaged eggs more distinguishable. By refining the preprocessing pipeline, we can further enhance the feature extraction process and model accuracy, leading to more reliable predictions. These advancements, combined with the ongoing optimization of deep learning architectures, will lead to more efficient quality control processes in the food industry. Producers will be able to detect even the smallest cracks and defects, improving product quality and ensuring more consistent, reliable products for consumers. Ultimately, these developments will contribute to minimizing food waste and improving consumer satisfaction by ensuring that eggs of higher quality reach the market, free from undetected damage.

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AUTHOR CONTRIBUTIONS

Talha Alperen Cengel: Conceptualization; investigation; writing—original draft; methodology; validation; visualization; writing—review and editing; software. **Bunyamin Gencturk**: Conceptualization; investigation; writing—original draft; methodology; validation; writing—review and editing; visualization; software. **Elham Tahsin Yasin**: Conceptualization; investigation; writing—original draft; methodology; validation; writing—review and editing; visualization; software. **Muslume Beyza Yildiz**: Conceptualization; investigation; writing—original draft; methodology; validation; visualization; writing—review and editing. **Ilkay Cinar**: Conceptualization; investigation; writing—original draft; methodology; validation; visualization; writing—review and editing; supervision. **Murat Koklu**: Conceptualization; investigation; writing—original draft; validation; visualization; writing—review and editing; methodology; supervision.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Contacting the corresponding author Abdullah Khan Kakar or accessing the study's dataset through the Kaggle Data website; https://www.kaggle.com/datasets/abdullahkhanuet22/eggs-images-classification-damaged-or-not.

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REFERENCES

Al-doorı, S. K. S., Taspınar, Y. S., & Koklu, M. (2021). Distracted driving detection with machine learning methods by cnn based feature extraction. *International Journal of Applied Mathematics Electronics and Computers*, *9*(4), 116–121. https://doi.org/10.18100/ijamec.1035749

Al-Huseiny, M. S., & Sajit, A. S. (2021). Transfer learning with GoogLeNet for detection of lung cancer. *Indonesian Journal of Electrical Engineering and Computer Science*, *22*(2), 1078–1086. https://doi.org/10.11591/ijeecs.v22.i2.pp1078-1086

Ayadi, W., Elhamzi, W., Charfi, I., & Atri, M. (2021). Deep CNN for brain tumor classification. *Neural Processing Letters*, *53*, 671–700. https://doi.org/10.1007/s11063-020-10398-2

Botta, B., Gattam, S. S. R., & Datta, A. K. (2022). Eggshell crack detection using deep convolutional neural networks. *Journal of Food Engineering*, *315*, 110798. https://doi.org/10.1016/j.jfoodeng.2021.110798

Cinar, I. (2023). Detection of chicken diseases from fecal images with the pre-trained Places365-GoogLeNet model. In *2023 IEEE 12th international conference on intelligent data acquisition and advanced computing systems: Technology and applications (IDAACS)*. (pp. 752–758). IEEE, Dortmund, Germany. https://doi.org/10.1109/IDAACS58523.2023.10348829

draw.io. (2021). *Open Source Diagramming is Moving to Diagrams.Net, Slowly* [Computer software].draw.io.

Gencturk, B., Arsoy, S., Taspinar, Y. S., Cinar, I., Kursun, R., Yasin, E. T., & Koklu, M. (2024). Detection of hazelnut varieties and development of mobile application with CNN data fusion feature reduction-based models. *European Food Research and Technology*, *250*(1), 97–110. https://doi.org/10.1007/s00217-023-04369-9

Gurucharan, M. K. (2024). *Basic CNN Architecture: A detailed explanation of the 5 layers in convolutional neural networks*, Upgrade Publishing. https://www.upgrad.com/blog/basic-cnn-architecture/

Haoran, C., Chuchu, H., Minlan, J., & Xiaoxiao, L. (2020). Egg crack detection based on support vector machine. In *2020 international conference on intelligent computing and human-computer interaction (ICHCI)*. (pp. 80–83). IEEE, Sanya, China. https://doi.org/10.1109/ICHCI51889.2020.00025

Huang, L., He, A., Zhai, M., Wang, Y., Bai, R., & Nie, X. (2019). A multi-feature fusion based on transfer learning for chicken embryo eggs classification. *Symmetry*, *11*(5), 606. https://doi.org/10.3390/sym11050606

Hussain, M., Bird, J. J., & Faria, D. R. (2019). A study on CNN transfer learning for image classification. In *Advances in computational intelligence systems: Contributions presented at the 18th UK workshop on computational intelligence* (pp. 191–202). Springer.

Isik, H., Tasdemir, S., Taspinar, Y. S., Kursun, R., Cinar, I., Yasar, A., Yasin, E. T., & Koklu, M. (2024). Maize seeds forecasting with hybrid directional and bi-directional long short-term memory models. *Food Science & Nutrition*, *12*(2), 786–803. https://doi.org/10.1002/fsn3.3783

Junaidi, A., Adhinata, F. D., Iskandar, A. R., & Lasama, J. (2022). Image classification for egg incubator using transfer learning VGG16 and InceptionV3. In *Proceedings of the 2nd international conference on electronics, biomedical engineering, and health informatics: ICEBEHI 2021*. (pp. 85–95). Springer. https://doi.org/10.1007/978-981-19-1804-9_7

Kakar, A. K. (2024). *Egg image dataset*. Kaggle. https://www.kaggle.com/ds/4361124

Kaya, Y., & Gürsoy, E. (2023). A MobileNet-based CNN model with a novel fine-tuning mechanism for COVID-19 infection detection. *Soft Computing*, *27*(9), 5521–5535. https://doi.org/10.1007/s00500-022-07798-y

17503841, 2025, 1, Downloaded from https://ift.onlinelibrary.wiley.com/doi/10.1111/1750-3841.17553 by Cardiff Metropolitan, Wiley Online Library on [26/02/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Lai, C.-C., Li, C.-H., Huang, K.-J., & Cheng, C.-W. (2021). Duck eggshell crack detection by nondestructive sonic measurement and analysis. *Sensors*, *21*(21), 7299. https://doi.org/10.3390/s21217299

Mascarenhas, S., & Agarwal, M. (2021). A comparison between VGG16, VGG19 and ResNet50 architecture frameworks for Image Classification. In *2021 International conference on disruptive technologies for multi-disciplinary research and applications (CENTCON)*. (pp. 96–99). IEEE, Bengaluru, India. https://doi.org/10.1109/CENTCON52345.2021.9687944

Payne, J. (2021). *Diagrams.net—An essential tool for techies*. https://www.theserpent.co.uk/posts/diagrams.net-an-essential-tool-for-techies/

Sarwinda, D., Paradisa, R. H., Bustamam, A., & Anggia, P. (2021). Deep learning in image classification using residual network (ResNet) variants for detection of colorectal cancer. *Procedia Computer Science*, *179*, 423–431. https://doi.org/10.1016/j.procs.2021.01.025

Soltani, M., & Omid, M. (2015). Detection of poultry egg freshness by dielectric spectroscopy and machine learning techniques. *LWT-Food Science and Technology*, *62*(2), 1034–1042. https://doi.org/10.1016/j.lwt.2015.02.019

Taspinar, Y. S., & Cinar, I. (2023). Detection of broken eggs with deep CNNs and performance analysis of models. In *2023 IEEE 12th international conference on intelligent data acquisition and advanced computing systems: Technology and applications (IDAACS)*. (pp. 724–728). IEEE,Dortmund, Germany. https://doi.org/10.1109/IDAACS58523.2023.10348756

Theckedath, D., & Sedamkar, R. (2020). Detecting affect states using VGG16, ResNet50 and SE-ResNet50 networks. *SN Computer Science*, *1*(2), 79. https://doi.org/10.1007/s42979-023-02168-3

Thipakorn, J., Waranusast, R., & Riyamongkol, P. (2017). Egg weight prediction and egg size classification using image processing and machine learning. In *2017 14th international conference on electrical engineering/electronics, computer, telecommunications and information technology (ECTI-CON)*. (pp. 477–480). IEEE, Phuket, Thailand. https://doi.org/10.1109/ECTICon.2017.8096278

Tutuncu, K., Cinar, I., Kursun, R., & Koklu, M. (2022). Edible and poisonous mushrooms classification by machine learning algorithms. In *2022 11th Mediterranean conference on embedded computing (MECO)*. (pp. 1–4). IEEE, Budva, Montenegro. https://doi.org/10.1109/MECO55406.2022.9797212

Vasileva, A., Gorbunova, E., Vasilev, A., Peretyagin, V., Chertov, A., & Korotaev, V. (2018). Assessing exterior egg quality indicators using machine vision. *British Poultry Science*, *59*(6), 636–645. https://doi.org/10.1080/00071668.2018.1523535

Waranusast, R., Intayod, P., & Makhod, D. (2016). Egg size classification on android mobile devices using image processing and machine learning. In *2016 Fifth ICT international student project conference (ICT-ISPC)*. (pp. 1–4). IEEE, Nakhonpathom, Thailand.

Wen, L., Li, X., Li, X., & Gao, L. (2019). A new transfer learning based on VGG-19 network for fault diagnosis. In *2019 IEEE 23rd international conference on computer supported cooperative work in design (CSCWD)*. (pp. 205–209). IEEE, Porto, Portugal. https://doi.org/10.1109/CSCWD.2019.8791884

Wu, L., Wang, Q., Jie, D., Wang, S., Zhu, Z., & Xiong, L. (2018). Detection of crack eggs by image processing and soft-margin support vector machine. *Journal of Computational Methods in Sciences and Engineering*, *18*(1), 21–31.

Yasar, A., Saritas, I., Sahman, M., & Cinar, A. (2019). Classification of Parkinson disease data with artificial neural networks. *IOP Conference Series: Materials Science and Engineering*, *675*, 012031. https://doi.org/10.1088/1757-899X/675/1/012031

Yasin, E. T., Kursun, R., & Koklu, M. (2023). Deep learning-based classification of black gram plant leaf diseases: A comparative study. In *11th International conference on advanced technologies (ICAT'23)*. (pp. 1–8). Istanbul, Turkiye. https://doi.org/10.58190/icat.2023.9

Yasin, E. T., Ozkan, I. A., & Koklu, M. (2023). Detection of fish freshness using artificial intelligence methods. *European Food Research and Technology*, *249*(8), 1979–1990. https://doi.org/10.1007/s00217-023-04271-4

Zaki, S. Z. M., Zulkifley, M. A., Stofa, M. M., Kamari, N. A. M., & Mohamed, N. A. (2020). Classification of tomato leaf diseases using MobileNet v2. *IAES International Journal of Artificial Intelligence*, *9*(2), 290. https://doi.org/10.11591/ijai.v9.i2.pp290-296

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